Setting up and running CpHMD

Prerequisites

charmm

amber

chamber (parmed)

see also github.com/dkortzak/CpHMD

Force-fields and engines

If you are familiar with GROMACS you know the difference between engine and force-field

CHARMM and AMBER both are names for force-field as well as MD engines

Here we use the CHARMM **force-field** and AMBER **engine**

AMBER was never designed to do that, but there are workarounds:

First generate CHARMM input files and then convert them into AMBER input files with **parmed**

CHARMM CpHMD force-field files

All potential protons have to present as a (dummy) particles

For HIS just take HSP

For ASP and GLU we need a residue with (dummy) protons on both oxygens

AMBER vs GROMACS vs CHARMM

GROMACS and AMBER are similar in having tons of binaries for different tasks

pdb2gmx - tleap (only works with AMBER force-field)

grompp + mdrun - pmemd

trjconv, trjcat, rms, rmsf, … are all collected in **cpptraj** (more details tomorrow)

CHARMM is one single binary for everything. What it does is defined in the input file.

AMBER vs GROMACS vs CHARMM

mdin vs mdp files

Overall very similar, parameter values are more are cryptic in AMBER and more self-explanatory in GROMACS

Whereas grompp generates a single input file with all settings stored in the tpr file, AMBER skips this step. Topology and parameters are directly fed to the MD executable.

AMBER GPU implementation

CPU is not used at all except for i/o

Only one GPU should be used per system (it can use multiple but performance gain is bad)

If you have multiple GPUs per node you should start several systems in parallel, e.g. different pH values

Makes it very cost efficient but not on a cluster which was designed with GROMACS in mind

CpHM workflow

1. Start with a pdb file of the (pre-equilibrated) system
2. Separate components
3. Separate big files (only 100,000 atoms per file)
4. Renumber residues and atoms (always start from 1 in each file)
5. Adjust residue names
6. Remove all protons from protein
7. generate charmm coordinate and topology files for each component
8. assemble them back into one coordinate and one topology file
9. convert those to amber input files
10. Run CpHMD with AMBER engine

CpHMD workflow

1. Start with a pdb file of the (pre-equilibrated) system
2. Separate components (including protein chains if there is more than 1)
3. Remove all protons from proteins
4. All components should have segid (columns 73-76) matching the charmm input file
5. Separate big files (only 100,000 atoms per file, do not break molecules)
6. Renumber residues and atoms (always start from 1 in each file)
7. adjust residue names
8. make sure there is a “TER” at end of each file

Many ways to achieve this, do whatever you prefer.

E.g. grep, sed and some python scripts, gmx editconf

Residue names to adjust

all aspartates need to be named ASP

all glutamates need to be named GLU

all histidines need to be named HSP

all waters need to be named TIP3

These are just the standard residue names, you might add your own residues if you want to mimic pH gradients or target parameters of a specific residue

Structure processing example

```
grep PROA system.pdb > proah.pdb
grep POPC system.pdb > memb.pdb
grep TIP3 system.pdb > bwat.pdb
grep CLA system.pdb > cla.pdb
grep SOD system.pdb > sod.pdb

sed -i '/TER  /d' memb.pdb
sed -i '/TER  /d' bwat.pdb
sed -i 's/OH2 TIP /OH2 TIP3/g' bwat.pdb
sed -i 's/H1  TIP /H1  TIP3/g' bwat.pdb
sed -i 's/H2  TIP /H2  TIP3/g' bwat.pdb
sed -i 's/  0.00           /  0.00      TIP3 /g' bwat.pdb
sed -i "s/TIP3S/TIP3 /g" bwat.pdb
sed -i "s/SOLV/TIP3 /g" bwat.pdb
python noh.py

sed -i 's/HSE/HSP/g' proa.pdb
sed -i 's/HSD/HSP/g' proa.pdb
sed -i '/H.*HSP/d' proa.pdb


python sepwat.py
python sepmemb.py
python sepprot.py
python sepions.py
```

CHARMM input files (non-protein)

```
stream toppar.str

open unit 1 read form name sod.pdb
read sequ pdb unit 1
gene IONS setup warn first none last n

open unit 1 read card name sod.pdb
read coor pdb unit 1 resid

IOFOrmat EXTEnded

open write unit 1 card name sod1.psf
write psf  unit 1 card

open write card unit 1 name sod1.crd
write coor unit 1 card

stop
```

Load force-field

Load pdb

generate and save topology and coordinates

CHARMM input files

```
set input prob.pdb
set outname prob_h
set segid PROB
! read in modified top and par files
set TOPPAR = toppar
set top = @TOPPAR/top_all22_prot.rtf
set par = @TOPPAR/par_all22_prot.prm
set watstr = @TOPPAR/toppar_water_ions.str
set phstr = @TOPPAR/toppar_phmd_c22_foramber.str

open unit 1 read form name @top
read rtf card unit 1
close unit 1

open unit 1 read form name @par
read param card unit 1 flex
close unit 1

stream @phstr
stream @watstr

! read in sequence
open unit 1 read form name @input
read sequ pdb unit 1

! generate psf
generate @segid first ACE last CT3 setup
!generate @segid
```

```
! patch dummy atoms for ASP and GLU
! ---------------------------------
define ANY sele bynu 1 end
set res1 = ?selresi          ! get the starting value of resid
calc reslast = ?selresi + ?nres -1

set ires @res1

prnlev -1 !turn off printing

label patch
 define residue sele segid @segid .and. resid @ires show end
 if ?selresn eq asp patch aspp2 @segid @ires
 if ?selresn eq glu patch glup2 @segid @ires
 incr ires
if ires le @reslast goto patch

autogen angles dihed

hbuild sele segi @segid .and. resn asp .and. type HD* end ! Build coordinates for dummy hydrogen in Asp
hbuild sele segi @segid .and. resn glu .and. type HE* end ! Build coordinates for dummy hydrogen in Glu

prnlev 5

! Edit Dihedrals for Glupp2 and Aspp2
!------------------------------------

define ANY sele bynu 1 end
set res1 = ?selresi          ! get the starting value of resid
calc reslast = ?selresi + ?nres -1

ic gene
ic fill
```

not needed if no ASP or GLU in system

CHARMM input files

```
! generate psf
generate @segid first ACE last CT3 setup
!generate @segid
```

Cap termini, if not wanted use other line:

generate @segid

parmed

```
chamber -top toppar/top_all22_prot.rtf -top toppar/top_all36_na.rtf -top toppar/top_all36_lipid.rtf -top toppar/top_all36_carb.rtf -top toppar/top_all36_cgenff.rtf -param toppar/par_all22_prot.prm -param toppar/par_all36_na.prm -param toppar/par_all36_carb.prm -param toppar/par_all36_lipid.prm -param toppar/par_all36_cgenff.prm -str toppar/toppar_water_ions.str -str toppar/toppar_phmd_c22_foramber.str -psf hhv1.psf -crd hhv1.crd -box 83.13,83.13,103.25,90,90,90

parmout hhv1.parm7 hhv1.rst7

go
```

Adjust filenames and box vectors (can find them in assemble_all output)

parmed Hydrogen mass repartitioning

We can redistribute some masses to H atoms and then use a 4fs time-step.

```
parm hhv1.parm7
hmassrepartition
outparm hhv1_0.parm7
quit
```

pmemd arguments

-O overwrite outputs

unlike GROMACS, AMBER will not make backups of files that already exists.

In case an output file already exists, it will either crash if -O is not set or overwrite without asking or backing up.

pmemd arguments inputs

-i MD parameter file

-p Topology

-c Start structure or restart file (may have positions only or positions and velocities)

-ref Reference for position restraints

CpHMD specific files

-phmdparm CpHMD parameters

-phmdin phmd settings

-phmdstrt state of lambda variables

pmemd arguments outputs

-r Restart file

-x trajectory

-o logfile

logfile contains lots of information (gromacs would save in ener.edr)

CpHMD specific files

-phmdout protonation states

-phmdrestrt Restart file for CpHMD

Step1 Minimization

Done without CpHMD activated

Safer to do on CPU

```
pmemd -O -i mini.mdin -c input.rst -p topol.parm -ref input.rst -r em.rst -o em.log -x em.nc
```

Step2 Equilibrations

Several stages with different position restraints

Where I come from:

1. Position restraints on everything but water
2. Position restraints on all protein atoms
3. Position restraints on all protein backbone atoms

1 and 2 can be relatively short and fixed duration

3 should be done until it is done, how long that takes will be different for different systems

Can be done at one pH and then simulations at other pH values can be started from the equilibrated system

Just recommendation what you do for your system might be different

Step2 Equilibrations

pmemd.cuda -O -p input.parm -c em.rst -ref em.rst -r equi1.rst -x equi1.nc -o equi1.log -phmdparm charm22_cphmd.parm -phmdin phmdin_start -phmdrestrt equi1.phmdrst -phmdout equi1.lambda

After it finished first line of equi1.phmdrst has to be edited: change PHMDRST to PHMDSTRT

pmemd.cuda -O -p input.parm -c equi1.rst -ref em.rst -r equi2.rst -x equi2.nc -o equi2.log -phmdparm charm22_cphmd.parm -phmdin phmdin_restart -phmdrestrt equi2.phmdrst -phmdout equi2.lambda -phmdstrt equi1.phmdrst

Some AMBER parameters

```
Minimization input file in explicit solvent
 &cntrl
    ! Minimization options
    imin=1,         ! Turn on minimization
    maxcyc=500,    ! Maximum number of minimization cycles
    ncyc=500,      ! 100 steepest-descent steps, better for strained systems

    ! Potential energy function options
    cut=12.0,       ! nonbonded cutoff, in angstroms
    fswitch=10.0,   ! Force-based switching

    ! Control how often information is printed to the output file
    ntpr=100,       ! Print energies every 100 steps
    ntxo=2,         ! Write NetCDF format

    ! Restraint options
    ntr=1,          ! Positional restraints for proteins, sugars, ligands, and lipid head groups
    nmropt=1,       ! Dihedral restraints for sugars and lipids

    ! Set water atom/residue names for SETTLE recognition
    watnam='TIP3',  ! Water residues are named WAT
    owtnm='OH2',     ! Water oxygens are named O
 /
```

Some AMBER parameters

```
A NPT simulation for common production-level simulations
 &cntrl
    imin=0,        ! No minimization
    irest=1,       ! This IS a restart of an old MD simulation
    ntx=5,         ! So our inpcrd file has velocities

    ! Temperature control
    ntt=3,         ! Langevin dynamics
    gamma_ln=1.0,  ! Friction coefficient (ps^-1)
    temp0=310,   ! Target temperature

    ! Potential energy control
    cut=12.0,      ! nonbonded cutoff, in angstroms
    fswitch=10.0,  ! Force-based switching

    ! MD settings
    nstlim=50000, ! 250K steps, 500 ps total
    dt=0.004,      ! time step (ps)
```

4fs after hydrogen mass repartitioning
use 2fs without it

Some AMBER parameters

```
    ! Control how often information is printed
    ntpr=1000,       ! Print energies every 1000 steps
    ntwx=50000,       ! Print coordinates every 5000 steps to the trajectory
    ntwr=10000,      ! Print a restart file every 10K steps (can be less frequent)
!   ntwv=-1,         ! Uncomment to also print velocities to trajectory
!   ntwf=-1,         ! Uncomment to also print forces to trajectory
    ntxo=2,          ! Write NetCDF format
    ioutfm=1,        ! Write NetCDF format (always do this!)

    ! Wrap coordinates when printing them to the same unit cell
    iwrap=1,

    ! Constant pressure control.
    barostat=2,      ! MC barostat... change to 1 for Berendsen
    ntp=3,           ! 1=isotropic, 2=anisotropic, 3=semi-isotropic w/ surften
    pres0=1.0,       ! Target external pressure, in bar

    ! Constant surface tension (needed for semi-isotropic scaling). Uncomment
    ! for this feature. csurften must be nonzero if ntp=3 above
    csurften=3,      ! Interfaces in 1=yz plane, 2=xz plane, 3=xy plane
    gamma_ten=0.0,  ! Surface tension (dyne/cm). 0 gives pure semi-iso scaling
    ninterface=2,    ! Number of interfaces (2 for bilayer)
```

better use barostat=1 for equilibration

this is for membrane systems, use ntp=1 if there is none

this is for membrane systems, delete if there is none

Some AMBER parameters

3 means explicit solvent

the pH for the simulation

```
iphmd = 3, solvph = 7.0,
efx = 0, efy = 0, efz = 0.0583,
```

settings for voltage, set all to 0 if there should be not voltage

Position restraints

Two ways to define restraints:

```
! Restraint options
ntr=1,         ! Positional restraints for proteins, sugars, ligands, and lipid head groups
restraintmask="@CA,N,C",
restraint_wt=1.0,
```

```
Protein posres
10.0
RES 1 134
END
Membrane posres
10.0
FIND
P * * POPC
SEARCH
RES 1 10946
END
END
```

Selection syntax

Syntax	Example
:{residue numlist}	':1-10', ':1,3,5', ':1-3,5,7-9'
@{atom numlist}	'@12,17', '@54-85', '@12,54-85,90'
:{residue namelist}	':LYS', ':ARG,ALA,GLY'
@{atom namelist}	'@CA', '@CA,C,O,N,H'
@%{atom type name}	'@%CT'
@/{atom_element_name}	'@/N'

Selection for CHAIN ID's	#Select chain A
	`::A`
	# Select residues 1-10 of chains A, B, and C
	`::A,B,C:1-10`
	# Select protein backbone atoms of residues 2-20 in chains A and B
	`::A,B:2-20@CA,C,O,N`

Selection syntax

:ALA,TRP	All alanine and tryptophan residues.
:5,10@CA CA	carbon in residues 5 and 10.
:*&!@H=	All non-hydrogen atoms (equivalent to “!@H=”).
@CA,C,O,N,H	All (protein) backbone atoms.
!@CA,C,O,N,H	All non-backbone atoms (=sidechains for proteins only).
:1-500@O&!(:WAT|:LYS,ARG)	All backbone oxygens in residues 1-500 but not in water, lysine or arginine residues.
(:11@CD<:5.5)&:Na+	Select all residues within 5.5 Ang. of atom CD from residue 11) AND those residues must be named Na+

CpHMD parameters

Here can target different residues

```
&phmdparm
  NGT = 13,
  NUMCH(:) = 14,17,18,22,11,4,3,14,17,18,22,11,14
  RES_NAME(:) = 'ASP','GLU','HSP','LYS','CYS','TIPP', 'TIPU','AST','GLT','HST','LYT','CYT','ASB',
  RES_TYPE(:) = 4,4,2,0,0,-2,-2,4,4,2,0,0,4,
  ATOM_NAME(1,:) = 'N','HN','CA','HA','CB','HB1','HB2','CG','OD1','HD1','OD2','HD2','C','O',
  ATOM_NAME(2,:) = 'N','HN','CA','HA','CB','HB1','HB2','CG','HG1','HG2','CD','OE1','HE1','OE2','HE2','C','O',
  ATOM_NAME(3,:) = 'N','HN','CA','HA','CB','HB1','HB2','CD2','HD2','CG','NE2','HE2','ND1','HD1','CE1','HE1','C','O',
  ATOM_NAME(4,:) = 'N','HN','CA','HA','CB','HB1','HB2','CG','HG1','HG2','CD','HD1','HD2','CE','HE1','HE2','NZ','HZ1','HZ2','HZ3','C','O',
  ATOM_NAME(5,:) = 'N','HN','CA','HA','CB','HB1','HB2','SG','HG1','C','O',
  ATOM_NAME(6,:) = 'OH2', 'H1', 'H2', 'H3',
  ATOM_NAME(7,:) = 'OH2', 'H1', 'H2',
  ATOM_NAME(8,:) = 'N','HN','CA','HA','CB','HB1','HB2','CG','OD1','HD1','OD2','HD2','C','O',
  ATOM_NAME(9,:) = 'N','HN','CA','HA','CB','HB1','HB2','CG','HG1','HG2','CD','OE1','HE1','OE2','HE2','C','O',
  ATOM_NAME(10,:) = 'N','HN','CA','HA','CB','HB1','HB2','CD2','HD2','CG','NE2','HE2','ND1','HD1','CE1','HE1','C','O',
  ATOM_NAME(11,:) = 'N','HN','CA','HA','CB','HB1','HB2','CG','HG1','HG2','CD','HD1','HD2','CE','HE1','HE2','NZ','HZ1','HZ2','HZ3','C','O',
  ATOM_NAME(12,:) = 'N','HN','CA','HA','CB','HB1','HB2','SG','HG1','C','O',
  ATOM_NAME(13,:) = 'N','HN','CA','HA','CB','HB1','HB2','CG','OD1','HD1','OD2','HD2','C','O',
  CH(1,:) = -0.47,0.31,0.07,0.09,-0.21,0.09,0.09,0.75,-0.61,0.44,-0.55,0,0.51,-0.51,-0.47,0.31,0.07,0.09,-0.21,0.09,0.09,0.75,-0.55,0,-0.61,0
  CH(2,:) = -0.47,0.31,0.07,0.09,-0.18,0.09,0.09,-0.21,0.09,0.09,0.75,-0.61,0.44,-0.55,0,0.51,-0.51,-0.47,0.31,0.07,0.09,-0.18,0.09,0.09,-0.2
```

CpHMD parameters

Here can target different residues

```
&phmdparm
  NGT = 13,
  NUMCH(:) = 14,17,18,22,11,4,3,14,17,18,22,11,14
  RES_NAME(:) = 'ASP','GLU','HSP','LYS','CYS','TIPP', 'TIPU','AST','GLT','HST','LYT','CYT','ASB',
  RES_TYPE(:) = 4,4,2,0,0,-2,-2,4,4,2,0,0,4,
```

In this example:
AST, GLT, HST, LYT, CYT get different pKa value assigned while all other parameters are unchanged, i.e. they ‘feel’ a different pH

$$U^{\mathrm{pH}}(\lambda_\alpha) = ln10 \cdot k_{\mathrm{B}}T\left(\mathrm{pH} - pK_a^{\mathrm{mod}}\right)\lambda_\alpha$$

CpHMD parameters

```
&phmdparm
  NGT = 13,
  NUMCH(:) = 14,17,18,22,11,4,3,14,17,18,22,11,14
  RES_NAME(:) = 'ASP','GLU','HSP','LYS','CYS','TIPP', 'TIPU','AST','GLT','HST','LYT','CYT','ASB',
  RES_TYPE(:) = 4,4,2,0,0,-2,-2,4,4,2,0,0,4,
```

Here can target different residues individually

In this example:
ASB is Aspartate with an increased value for the barrier (in the system I was simulating one particular ASP consistently sampled intermediate protonation states)

CpHMD parameters

```
&phmdparm
  NGT = 13,
  NUMCH(:) = 14,17,18,22,11,4,3,14,17,18,22,11,14
  RES_NAME(:) = 'ASP','GLU','HSP','LYS','CYS','TIPP', 'TIPU','AST','GLT','HST','LYT','CYT','ASB',
  RES_TYPE(:) = 4,4,2,0,0,-2,-2,4,4,2,0,0,4,
```

Here can target different residues individually

In both cases, only CpHMD parameters are changed and the only files which needs to be modified is the parameter file as well as topology **post-processing**.

There might be cases where you only want certain residues to be titratable and you would need to make copies in the force-field **pre-processing** to prevent hydrogens to be added to non-titratable residues.

Modifications to mimic pH gradients

1. Make a copy of the parameter in the input file

2. Edit the pKa of the copied residue

3. Change all residues that should see a different pH in the parm7 and pdb file to the new residue type

$$U^{\mathrm{pH}}(\lambda_\alpha) = ln10 \cdot k_{\mathrm{B}}T\left(\mathrm{pH} - pK_a^{\mathrm{mod}}\right)\lambda_\alpha$$

Target barrier of individual residues

Overall same approach as before

Only difference: Edit barrier parameter in the input file

CpHMD settings

```
&phmdin
    QMass_PHMD = 10,                                   ! Mass of the Lambda Particle
    Temp_PHMD = 310,                                   ! Temp of the lambda particle
    phbeta = 5,                                        ! Friction Coefficient (1/ps) for titration integrator
    iphfrq = 1,                                        ! Frequency to update the lambda forces
    NPrint_PHMD = 5000,                                 ! How often to print the lambda
    PrLam = .true.,                                    ! Should lambda be printed
    PrDeriv = .false.,                                 ! Do you want to print the derivatives?
    PRNLEV = 7,                                        ! Sets what is printed out, (7) normal print level
    PHTest = 0,                                        ! Are lambda and theta fixed, (0) = not fixed, (1) = fixed
    MaskTitrRes(:) = 'ASP','GLU','HSP','AST','GLT','HST',  ! Residues to include as titratable
    MaskTitrResTypes = 6,                              ! number of titratable residues
    QPHMDStart = .true.,                               ! Initialize velocities of titration varibles with a Boltzmann distribution
/
```

should match the residue types from the parameter file

CpHMD outputs

*.nc: trajectories

*.lambda: the protonation states for all titratable residues.

lambda files

First column: No. of steps (not time)

Then two columns for each residue:

1. protonation state
2. tautomer state

tautomer only meaningful for protonated ASP,GLU or de-protonated HIS

lambda files

Standard conventions:

$\lambda < 0.2$: protonated

$\lambda > 0.8$ de-protonated

Intermediate values are unphysical

First thing to check: λ should not sample too many intermediates. If there is a residue with significant amount of $0.2 < \lambda < 0.8$ the barrier of that residue should be adjusted

titration curves

Plot fraction de-protonated (S) vs pH and fit this equation:

$$S = \frac{1}{1 + 10^{n(\mathrm{p}K_a - \mathrm{pH})}}$$

Note: The n in the equation is a Hill-coefficient and should be 1 in most cases. For a single proton binding to one amino-acid it does not really make sense to include it.

Some people like to include it but it's not really physical and unclear how to interpret it.

see also

▸ J Chem Biol. 2009 Sep 25;3(1):37–44. doi: 10.1007/s12154-009-0029-3

Hill coefficients, dose–response curves and allosteric mechanisms

Heino Prinz [1,✉]

Not much going on, just a 'standard' ligand binding curve with $[H]=10^{-pH}$

cpptraj

does what gmx trjcat, tjrconv, … does in one program that reads scripts

example: (put everything in box and fit to reference structure)

parm hhv1.parm7

trajin seg*.nc

reference hhv1.pdb

autoimage

align :1-134@CA,C,O,N reference

trajout fit.nc

go

Notes about charges

We started with neutral systems for fixed protonation

Changing protonation state will change the net charge

PME will compensate with a uniform background charge

This might give some artifacts in regions where there should be no charge, e.g. inside membranes or protein interiors.

Notes about charges

Solutions to this problem include:

Add titratable water molecules

Add different amount of counter ions at different pHs

Increase system size, the larger the system the smaller the background charge density

See also

▸ J Phys Chem B. Author manuscript; available in PMC: 2025 Jul 7.

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Force Field Limitations of All-Atom Continuous Constant pH Molecular Dynamics

Craig A Peeples [1], Ruibin Liu [1], Jana Shen [1]

› J Chem Theory Comput. 2016 Nov 8;12(11):5411-5421. doi: 10.1021/acs.jctc.6b00552. Epub 2016 Oct 24.

All-Atom Continuous Constant pH Molecular Dynamics With Particle Mesh Ewald and Titratable Water

Yandong Huang [1], Wei Chen [1], Jason A Wallace [2], Jana Shen [1]